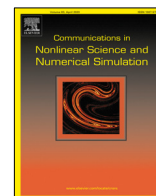




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Research paper

Observability and observer design for a class of hyperbolic PDEs with van de Pol type boundary conditions

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ABSTRACT

This paper focuses on observability and observer design for nonlinear complex dynamical systems described by a class of hyperbolic partial differential equations (PDEs) with nonlinear van de Pol type boundary conditions. The systems exhibit complex dynamics due to its imbalance of energy flows. Both the exact observability and approximate observability of the systems with different boundary output measurements are shown by using methods of characteristics and boundary nonlinear reflections. Motivated by the approximate observability of the systems, a PDE state observer by using the boundary displacement measurement only is designed, and a sufficient condition to guarantee the estimation error systems to be exponentially stable is given. Theoretical results are proved rigorously, with some numerical simulations performed to validate the effect of the proposed observer.

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1. Introduction

In practice, obtaining present state of dynamical systems is momentous either to design controller or to acquire real-time information on the systems for decision-making or monitoring [1]. In most scenarios not all the states are directly measurable because placing all the necessary sensors to measure states may not be feasible physically or the costs may be expensive. In such cases, the unmeasurable states can be estimated usually by use of state observers, and an efficient observer can exist only if some observability or detectability properties are satisfied. Observability and observer design for continuous-time dynamical systems have been the vital topics in the field of control theory. Observability and observer design for lumped parameter systems have been receiving extensive attentions, see, e.g., [1–4], and the counterpart for distributed parameter systems described by partial differential equations (PDEs) becomes more intricate and attracting when the observation is at the boundaries only.

In applications, there are many processes that can be described by hyperbolic PDEs, such as chemical reactors [5], population [6], traffic flow [7], and blood flow [8]. Observability and observer design for hyperbolic PDEs has attracted many interests due to its potential applications in secure communications, biological systems, and quantum mechanics [9–12], among many others. For example, Zuazua addressed several topics concerning the observation and control of wave propagation phenomena modeled by finite difference methods in [13]; Fridman applied a linear matrix inequality method in the observer design and the initial state recovering by iterative forward and backward observers for a class of semilinear systems described by wave and beam equations with boundary measurements from a finite interval in [14];

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Feng and Guo studied observer design for a 1-D wave equation with linear boundary condition by boundary displacement measurement only in [15]; Haak and Hoang analyzed exact observability of a 1-D wave equation with linear boundary condition on a noncylindrical domain in [16]. For more relevant progresses, see [17–20] and the references therein.

It is noteworthy that aforementioned studies on observability and observer design for 1-D linear hyperbolic PDEs are all under linear boundary conditions, while little research has been done for hyperbolic PDEs with nonlinear ones. To the best of authors' knowledge, in available literature only Cai and Xiao investigated observability of a 1-D wave equation with van der Pol type nonlinear boundary condition at the right-end in [2]. Motivated by the work in [2,15], in this paper one develops observability and observer design for a class of hyperbolic PDEs with van de Pol type nonlinear boundary condition at the left-end as follows:

$$\begin{cases} w_{tt} - w_{xx} + \gamma w_{xt} + 2kw_t + k\gamma w_x + k^2w = 0, & x \in (0, 1), t > 0, \\ w_x(0, t) = -\alpha[w_t(0, t) + kw(0, t)] + \beta[w_t(0, t) + kw(0, t)]^3, & t \geq 0, \\ w_x(1, t) = 0, & t \geq 0, \\ w(x, 0) = w_0(x), w_t(x, 0) = w_1(x), & x \in [0, 1], \end{cases} \quad (1.1)$$

where $\gamma, k \in \mathbb{R}$, $\beta > 0$, $\alpha < \gamma_1$ with $\gamma_1 = \frac{-\gamma + \sqrt{4 + \gamma^2}}{2}$, and $(w_0(x), w_1(x))$ is the initial state. The hyperbolic PDE in (1.1) can be seen as a wave equation when $k = 0$, and a telegraph equation [21] or a Klein–Gordon equation with a uniform distributed damping term [22] when $\gamma = 0$ and $k \neq 0$. In particular, Chen et al. [22] showed the chaotic behavior of (1.1) with $\gamma = 0$ and $k \neq 0$, where the van de Pol type boundary condition is at the right-end. In addition, one defines

$$H_{\perp}^1(0, 1) = \left\{ f \in H^1(0, 1) \mid \int_0^1 f dx = 0 \right\}, \quad (1.2)$$

then $H_{\perp}^1(0, 1)$ is a closed subspace of $H^1(0, 1)$ and it inherits the inner product and norm of $H^1(0, 1)$. Throughout this paper, one considers system (1.1) in the state space

$$\mathcal{H} = H_{\perp}^1(0, 1) \times L^2(0, 1) \quad (1.3)$$

with inner product

$$\langle (f_1, g_1), (f_2, g_2) \rangle_{\mathcal{H}} = \int_0^1 [f_1' \overline{f_2'} + (g_1 + kf_1) \overline{(g_2 + kf_2)}] dx. \quad (1.4)$$

The total energy of (1.1) in the state space $\mathcal{H}$ at the time t is defined to be

$$E(t) = \frac{1}{2} \int_0^1 [w_x^2 + (w_t + kw)^2] dx.$$

By the standard energy multiplier technique [22], the change rate of $E(t)$ is

$$\begin{aligned} \dot{E}(t) &= \left[w_x(w_t + kw) - \frac{\gamma}{2}(w_t + kw)^2 \right] \Big|_{x=0}^{x=1} - 2kE(t) \\ &= \underbrace{-\frac{\gamma}{2}(w_t(1, t) + kw(1, t))^2}_{\varepsilon_1} + \underbrace{\left(\alpha + \frac{\gamma}{2} - \beta[w_t(0, t) + kw(0, t)]^2 \right) (w_t(0, t) + kw(0, t))^2}_{\varepsilon_2} \\ &\quad - 2kE(t). \end{aligned} \quad (1.5)$$

Note that

$$\begin{cases} \varepsilon_1 \geq 0, & \text{if } \gamma < 0, \\ \varepsilon_1 \leq 0, & \text{if } \gamma > 0, \end{cases} \quad \begin{cases} \varepsilon_2 \geq 0, & \text{if } |w_t(0, t) + kw(0, t)| \leq \sqrt{\frac{2\alpha + \gamma}{2\beta}}, \\ \varepsilon_2 \leq 0, & \text{if } |w_t(0, t) + kw(0, t)| \geq \sqrt{\frac{2\alpha + \gamma}{2\beta}}, \end{cases} \quad \begin{cases} -kE(t) \geq 0, & \text{if } k < 0, \\ -kE(t) \leq 0, & \text{if } k > 0. \end{cases} \quad (1.6)$$

ε_1 and $-kE(t)$ can be considered as boundary antidamping/damping and distributed antidamping/damping, respectively; ε_2 can cause boundary energy regulation. Thus, it is seen that $\dot{E}(t)$ given by (1.5) is indefinite due to the imbalance of energy flows, which can lead to many complex dynamics, including bifurcation, homoclinic and heteroclinic orbits, and chaotic vibration when system parameters vary within certain ranges [23–27]. It is seen from (1.6) that the interaction between PDE and nonlinear boundary condition lead to the occurrence of complex dynamics of (1.1).

By setting two different boundary output measurements

$$\mathbf{y}_{obs}(t) = (w(0, t), w_t(0, t))^{\top} \quad \text{and} \quad y_{obs}(t) = w(0, t),$$

for different kinds of observability requirements, the exact observability and approximate observability of (1.1) will be studied, respectively. Compared with the studies on observability and observer design for 1-D wave equations with linear boundary conditions, the main difficulty and challenge caused by the nonlinear boundary condition in (1.1) are that the observation map from the space of initial states to the one of output measurement has low regularities, which leads to obstacle in the analysis of the inverse of the observability map. Thus, novel theoretical analysis approaches should

be developed additionally to obtain the stability of the estimation error systems. By using the methods of characteristic and two nonlinear reflection relations, some sufficient conditions concerning the exact observability and approximate observability of (1.1) will be developed. A PDE state observer is further designed by boundary displacement measurement output $w(0, t)$ only, and a sufficient condition is given to guarantee the exponential stability of the resulting estimation error systems. The main contribution and novelty of this paper are follows:

- (1) Compared with the previous exact observability and approximate observability for a wave equation with van de Pol type boundary condition [2], this paper develops both exact observability and approximate observability for a class of hyperbolic PDEs with van de Pol type nonlinear boundary conditions.
- (2) Different from the asymptotic convergence of the PDE state observer by using the measurement of boundary velocity [28], this paper presents the exponential convergence of the PDE state observer using boundary displacement measurement.
- (3) Different from the conventional use of Fourier analysis, operator semigroup, and Lyapunov direct approaches in most available literature (e.g., [15,16,19]), this paper develops the method of characteristics and nonlinear boundary reflection to estimate the observability constant and analyze the exponential convergence of the PDE state observer, respectively.

The contents of this paper are organized as follows. Section 2 first investigates the exact observability and approximate observability of system (1.1). In Section 3, a PDE state observer by making use of boundary displacement measurement $w(0, t)$ is designed, and a sufficient condition is developed to guarantee the expected convergence of the estimation errors. Section 4 addresses some numerical simulations to support the theoretical conclusions, followed up by some concluding remarks in Section 5. Throughout this paper, unless other stated, we omit the arguments x and t without confusion.

2. Observability of system (1.1)

In this section, for two different boundary output measurements $\mathbf{y}_{obs}(t) = (w(0, t), w_t(0, t))^T$ and $y_{obs}(t) = w(0, t)$, one gives the sufficient conditions to guarantee both the exact observability and approximate observability of system (1.1), respectively. The definitions of exact observability and approximate observability are introduced as follows.

Definition 2.1 ([2, Definition 2.2]). Given $T > 0$ and an output measurement $\mathbf{y}_{obs}(t)$, if the following inequality is true:

$$\int_0^T \|\mathbf{y}_{obs}(t)\|_{\mathbb{R}^2}^2 dt \geq C \|(w_0, w_1)\|_{\mathcal{H}}^2,$$

where $C > 0$ is a constant, then system (1.1) is said to be exactly observable on $[0, T]$ at the boundary $x = 0$.

Definition 2.2 ([29, Definition 2.3.6]). Given $T > 0$ and an output measurement $y_{obs}(t)$, if the map $(w_0, w_1) \in \mathcal{H} \rightarrow y_{obs}$ satisfying $\text{Ker}\{y_{obs}(t)\} = \{(0, 0)\}$, $\forall t \in [0, T]$, then system (1.1) is said to be approximately observable on $[0, T]$ at the boundary $x = 0$.

Remark 2.1. In the definitions of both exact observability and approximate observability, one is allowed to measure the output during the time interval $[0, T]$ at the boundary $x = 0$ and expects to recover complete information on the initial state (w_0, w_1) . Both of them also mean that the initial state can be uniquely determined by a given boundary output observation within a certain period of time, and it is obvious that exact observability implies approximate observability. For exactly observable system, the initial value can be reconstructed from output data with continuous dependence but the approximate observability cannot guarantee the continuous dependence yet has identifiability for initial value.

For the 1-D hyperbolic system (1.1), define two Riemann invariants (see [23,30])

$$u = \frac{1}{\gamma_1 + \gamma_2} [w_x + \gamma_1(w_t + kw)] \quad \text{and} \quad v = \frac{1}{\gamma_1 + \gamma_2} [w_x - \gamma_2(w_t + kw)], \quad (2.1)$$

for $x \in [0, 1]$ and $t \geq 0$, where γ_1 and γ_2 are determined by

$$\gamma_1 = \frac{-\gamma + \sqrt{4 + \gamma^2}}{2}, \quad \gamma_2 = \frac{\gamma + \sqrt{4 + \gamma^2}}{2}, \quad \gamma_2 - \gamma_1 = \gamma, \quad \text{and} \quad \gamma_1 \gamma_2 = 1. \quad (2.2)$$

Under (2.1), system (1.1) is transformed into the (u, v) system

$$\begin{cases} \frac{\partial}{\partial t} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} \gamma_1 & 0 \\ 0 & -\gamma_2 \end{bmatrix} \frac{\partial}{\partial x} \begin{bmatrix} u \\ v \end{bmatrix} - k \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}, & x \in (0, 1), \quad t > 0, \\ v(0, t) = G(u(0, t)), & t \geq 0, \\ u(1, t) = F(v(1, t)), & t \geq 0, \\ u(x, 0) = u_0(x), \quad v(x, 0) = v_0(x), & x \in [0, 1], \end{cases} \quad (2.3)$$

where

$$G(y) = y + g(y) \quad \text{and} \quad F(y) = -\gamma_1^2 y, \quad (2.4)$$

in which $g(y)$ is uniquely satisfying

$$\beta g^3(y) + (\gamma_1 - \alpha)g(y) + (\gamma_1 + \gamma_2)y = 0, \quad \alpha < 1, \quad \beta > 0, \quad (2.5)$$

and the initial state $(u_0, v_0) \in L^2(0, 1) \times L^2(0, 1)$ is

$$u_0(x) = \frac{1}{\gamma_1 + \gamma_2} [w'_0(x) + \gamma_1(w_1(x) + kw_0(x))], \quad v_0(x) = \frac{1}{\gamma_1 + \gamma_2} [w'_0(x) - \gamma_2(w_1(x) + kw_0(x))]. \quad (2.6)$$

Define

$$\mathcal{G}(\cdot) = G(e^{-\gamma_2 k \cdot}), \quad \mathcal{F}(\cdot) = F(e^{-\gamma_1 k \cdot}). \quad (2.7)$$

It follows from the linearity of $F(\cdot)$ in (2.4) that

$$F(e^{-(\gamma_1 + \gamma_2)k \cdot}) = e^{-\gamma_2 k \cdot} \mathcal{F}(\cdot).$$

Then, for $t = (\gamma_1 + \gamma_2)n + \tau$, $n \in \mathbb{N}$, $0 \leq \tau \leq \gamma_1 + \gamma_2$, and $x \in [0, 1]$, the solution (u, v) of (2.3) can be expressed explicitly as

$$u(x, t) = \begin{cases} e^{-\gamma_2 k(1-x)} (\mathcal{F} \circ \mathcal{G})^n (e^{-\gamma_2 k h_1(x, \tau)} u_0(h_1(x, \tau) + 1)), & 0 \leq \tau \leq \gamma_2(1-x), \\ e^{-\gamma_2 k(1-x)} \mathcal{G}^{-1} \circ (\mathcal{G} \circ \mathcal{F})^{n+1} (e^{-\gamma_1 k h_2(x, \tau)} v_0(-h_2(x, \tau))), & \gamma_2(1-x) < \tau \leq \gamma_2(1-x) + \gamma_1, \\ e^{-\gamma_2 k(1-x)} (\mathcal{F} \circ \mathcal{G})^{n+1} (e^{-\gamma_2 k h_3(x, \tau)} u_0(h_3(x, \tau) + 1)), & \gamma_2(1-x) + \gamma_1 < \tau \leq \gamma_1 + \gamma_2, \end{cases} \quad (2.8)$$

$$v(x, t) = \begin{cases} e^{-\gamma_1 k x} (\mathcal{G} \circ \mathcal{F})^n (e^{-\gamma_1 k h_4(x, \tau)} v_0(-h_4(x, \tau))), & 0 \leq \tau \leq \gamma_1 x, \\ e^{-\gamma_1 k x} \mathcal{G} \circ (\mathcal{F} \circ \mathcal{G})^n (e^{-\gamma_2 k h_5(x, \tau)} u_0(h_5(x, \tau) + 1)), & \gamma_1 x < \tau \leq \gamma_1 x + \gamma_2, \\ e^{-\gamma_1 k x} (\mathcal{G} \circ \mathcal{F})^{n+1} (e^{-\gamma_1 k h_6(x, \tau)} v_0(-h_6(x, \tau))), & \gamma_1 x + \gamma_2 < \tau \leq \gamma_1 + \gamma_2, \end{cases} \quad (2.9)$$

where

$$\begin{aligned} h_1(x, \tau) &= x + \gamma_1 \tau - 1, & h_2(x, \tau) &= \gamma_2^2 x + \gamma_2 \tau - \gamma_2^2 - 1, & h_3(x, \tau) &= x + \gamma_1 \tau - \gamma_1^2 - 2, \\ h_4(x, \tau) &= \gamma_2 \tau - x, & h_5(x, \tau) &= \gamma_1 \tau - \gamma_1^2 x - 1, & h_6(x, \tau) &= \gamma_2 \tau - x - \gamma_2^2 - 1. \end{aligned} \quad (2.10)$$

Lemma 2.1. Let $\alpha < \gamma_1$, $\beta > 0$, and $\gamma, k \in \mathbb{R}$. For the map $G : \mathbb{R} \rightarrow \mathbb{R}$ given by (2.4), there exists a sufficiently small constant $C_0 > 0$ such that

$$|G(y) - y| \geq C_0 |y|.$$

Moreover, $|G(y)| \leq K_0 |y|$, where $K_0 = \max\{\frac{\alpha + \gamma_2}{\gamma_1 - \alpha}, 1\}$.

Proof. Through the differential of (2.5) with respect to y , one has

$$g'(y) = -\frac{\gamma_1 + \gamma_2}{3\beta g^2(y) + \gamma_1 - \alpha} < 0,$$

which together with $g(0) = 0$, implies that $g(y) < 0$ for $y > 0$ and $g(y) > 0$ for $y < 0$. Then, for $y \neq 0$, it follows that

$$\left(\frac{g(y)}{y}\right)' = \frac{g'(y)y - g(y)}{y^2} = -\frac{2\beta g^3(y)}{y^2[3\beta g^2(y) + \gamma_1 - \alpha]} \begin{cases} > 0, & \text{if } y > 0, \\ < 0, & \text{if } y < 0. \end{cases}$$

Note that

$$\lim_{y \rightarrow 0} \frac{g(y)}{y} = \lim_{y \rightarrow 0} g'(y) = -\frac{\gamma_1 + \gamma_2}{\gamma_1 - \alpha} < 0 \quad \text{and} \quad \lim_{y \rightarrow \pm\infty} \frac{g(y)}{y} = 0.$$

Thus, it follows from $\alpha < \gamma_1$ that

$$0 < \left|\frac{g(y)}{y}\right| < \frac{\gamma_1 + \gamma_2}{\gamma_1 - \alpha},$$

which indicates that there exists a sufficiently small constant $C_0 > 0$ such that

$$|G(y) - y| = |g(y)| = \left|\frac{g(y)}{y}\right| |y| > C_0 |y|, \quad y \neq 0.$$

Simultaneously,

$$|G(y)| = |g(y) + y| = \left| \frac{g(y)}{y} + 1 \right| |y| \leq K_0 |y|, \quad \text{with } K_0 = \max \left\{ \left| \frac{\alpha + \gamma_2}{\gamma_1 - \alpha} \right|, 1 \right\}.$$

This ends the proof of [Lemma 2.1](#). $\square$

The exact observability and approximate observability of system (1.1) can be summarized as the following [Theorem 2.1](#).

Theorem 2.1. Let $\alpha < \gamma_1$, $\beta > 0$, and $\gamma, k \in \mathbb{R}$. For any $(w_0, w_1) \in \mathcal{H}$ and $T \geq \gamma_1 + \gamma_2$,

(i) if the boundary output measurement is given by

$$\mathbf{y}_{obs}(t) = (w(0, t), w_t(0, t))^T,$$

then system (1.1) is exactly observable on $[0, T]$ at the boundary $x = 0$;

(ii) if the boundary output measurement is given by

$$y_{obs}(t) = w(0, t),$$

then system (1.1) is approximately observable on $[0, T]$ at the boundary $x = 0$.

Proof. (i) According to (2.7), (2.8), and (2.9), one has for $0 \leq t \leq \gamma_1 + \gamma_2$,

$$u(0, t) = \begin{cases} e^{-kt} u_0(\gamma_1 t), & 0 \leq t \leq \gamma_2, \\ e^{-\gamma_2 k} F(e^{-k(t-\gamma_2)} v_0(1 + \gamma_2^2 - \gamma_2 t)), & \gamma_2 < t \leq \gamma_1 + \gamma_2, \end{cases}$$

and

$$v(0, t) = \begin{cases} G(e^{-kt} u_0(\gamma_1 t)), & 0 < t \leq \gamma_2, \\ G(e^{-\gamma_2 k} F(e^{-k(t-\gamma_2)} v_0(1 + \gamma_2^2 - \gamma_2 t))), & \gamma_2 < t \leq \gamma_1 + \gamma_2. \end{cases}$$

Therefore, for $0 \leq t \leq \gamma_2$,

$$u(0, t) - v(0, t) = e^{-kt} u_0(\gamma_1 t) - G(e^{-kt} u_0(\gamma_1 t)),$$

and for $\gamma_2 \leq t \leq \gamma_1 + \gamma_2$,

$$u(0, t) - v(0, t) = e^{-\gamma_2 k} F(e^{-k(t-\gamma_2)} v_0(1 + \gamma_2^2 - \gamma_2 t)) - G(e^{-\gamma_2 k} F(e^{-k(t-\gamma_2)} v_0(1 + \gamma_2^2 - \gamma_2 t))).$$

It then follows from [Lemma 2.1](#) and (2.4) that

$$\begin{aligned} \int_0^{\gamma_2} |u(0, t) - v(0, t)|^2 dt &= \int_0^{\gamma_2} |e^{-kt} u_0(\gamma_1 t) - G(e^{-kt} u_0(\gamma_1 t))|^2 dt \geq C_0^2 \gamma_2 \int_0^1 |e^{-k\gamma_2 t} u_0(t)|^2 dt \\ &\geq C_0^2 C_1 \int_0^1 |u_0(x)|^2 dx, \quad \text{with } C_1 = \gamma_2 \min\{1, e^{-2\gamma_2 k}\}, \end{aligned} \quad (2.11)$$

and

$$\begin{aligned} \int_{\gamma_2}^{\gamma_1 + \gamma_2} |u(0, t) - v(0, t)|^2 dt &\geq C_0^2 e^{-2\gamma_2 k} \int_{\gamma_2}^{\gamma_1 + \gamma_2} |F(e^{-k(t-\gamma_2)} v_0(1 + \gamma_2^2 - \gamma_2 t))|^2 dt \\ &\geq C_0^2 C_2 \int_0^1 |v_0(x)|^2 dx, \quad \text{with } C_2 = \gamma_1^5 e^{-2\gamma_2 k} \min\{1, e^{-2\gamma_1 k}\}. \end{aligned} \quad (2.12)$$

By (2.7), one has

$$\begin{aligned} \int_0^1 [|w'_0(x)|^2 + |w_1(x) + kw_0(x)|^2] dx &= \int_0^1 [|\gamma_2 u_0(x) + \gamma_1 v_0(x)|^2 + |u_0(x) - v_0(x)|^2] dx \\ &= \int_0^1 [(\gamma_2^2 + 1)|u_0(x)|^2 + (\gamma_1^2 + 1)|v_0(x)|^2] dx, \end{aligned}$$

which implies that

$$\int_0^1 [|u_0(x)|^2 + |v_0(x)|^2] dx \geq \frac{1}{C_3} \int_0^1 [|w'_0(x)|^2 + |w_1(x) + kw_0(x)|^2] dx, \quad (2.13)$$

where $C_3 = \max\{\gamma_1^2 + 1, \gamma_2^2 + 1\}$. Therefore, for $T \geq \gamma_1 + \gamma_2$, it follows from $w_t(x, t) + kw(x, t) = u(x, t) - v(x, t)$, (2.11), (2.12), and (2.13) that

$$\begin{aligned} \int_0^T |w_t(0, t) + kw(0, t)|^2 dt &\geq \int_0^{\gamma_1 + \gamma_2} |u(0, t) - v(0, t)|^2 dt \\ &\geq C_4 \int_0^1 (|u_0(x)|^2 + |v_0(x)|^2) dx \\ &\geq \frac{C_4}{C_3} \|(w_0, w_1)\|_{\mathcal{H}}^2, \end{aligned} \quad (2.14)$$

where $C_4 = C_0^2 \min\{C_1, C_2\}$. Hence, by (2.14), one obtains that

$$\begin{aligned} \int_0^T \|\mathbf{y}_{\text{obs}}(\mathbf{t})\|_{\mathbb{R}^2}^2 dt &= \int_0^T [|w(0, t)|^2 + |w_t(0, t)|^2] dt \\ &\geq \frac{1}{C_5} \int_0^T |w_t(0, t) + kw(0, t)|^2 dt \\ &\geq \frac{C_4}{C_3 C_5} \|(w_0, w_1)\|_{\mathcal{H}}^2, \end{aligned}$$

where $C_5 = 2 \max\{1, k^2\}$ and the observability constant can be found as $C = \frac{C_4}{C_3 C_5}$. This ends the proof of (i).

(ii) Note that

$$w_0(x) = \int_0^x [\gamma_2 u_0(s) + \gamma_1 v_0(s)] ds + w_0(0). \quad (2.15)$$

Since $u(0, t) = e^{-kt} u_0(\gamma_1 t)$ for $0 \leq t \leq \gamma_2$ and $u(0, t) - v(0, t) = w_t(0, t) + kw(0, t)$, it is seen that

$$\begin{aligned} \int_0^x \gamma_2 u_0(s) ds + w_0(0) &= \int_0^{\gamma_2 x} u_0(\gamma_1 s) ds + w_0(0) = \int_0^{\gamma_2 x} e^{ks} u(0, s) ds + w_0(0) \\ &= \int_0^{\gamma_2 x} e^{ks} [u(0, s) - v(0, s)] ds + w_0(0) + \int_0^{\gamma_2 x} e^{ks} v(0, s) ds \\ &= e^{k\gamma_2 x} w(0, \gamma_2 x) + \int_0^{\gamma_2 x} e^{ks} v(0, s) ds. \end{aligned} \quad (2.16)$$

By $v(0, t) = G(e^{-kt} u_0(\gamma_1 t))$ for $0 \leq t \leq \gamma_2$, $|G(y)| \leq K_0|y|$ in Lemma 2.1, and (2.11), one has

$$\begin{aligned} \int_0^{\gamma_2 x} |v(0, s)|^2 ds &= \int_0^{\gamma_2 x} |G(e^{-ks} u_0(\gamma_1 s))|^2 ds \leq K_0 K_1 \int_0^1 |u_0(s)|^2 ds \\ &\leq \frac{K_0 K_1}{C_0^2 C_1} \int_0^{\gamma_2} |u(0, t) - v(0, t)|^2 dt \leq \frac{2K_0 K_1}{C_0^2 C_1} \int_0^{\gamma_2} [|w_t(0, t)|^2 + k^2 |w(0, t)|^2] dt, \end{aligned} \quad (2.17)$$

where $K_1 = \gamma_2 \max\{1, e^{-2\gamma_2 k}\}$. Moreover, it follows from (2.12) that

$$\int_0^1 |v_0(s)|^2 ds \leq \frac{1}{C_0^2 C_2} \int_{\gamma_2}^{\gamma_1 + \gamma_2} [|w_t(0, t)|^2 + k^2 |w(0, t)|^2] dt. \quad (2.18)$$

Thus, using (2.15), $(a + b)^2 \leq 2(a^2 + b^2)$, (2.16), Hölder's inequality, (2.17), and (2.18), one obtains that

$$\begin{aligned} &\int_0^1 |w_0(x)|^2 dx \\ &= \int_0^1 \left| \int_0^x \gamma_2 u_0(s) ds + w_0(0) + \int_0^x \gamma_1 v_0(s) ds \right|^2 dx \\ &\leq 2 \int_0^1 \left| \int_0^x \gamma_2 u_0(s) ds + w_0(0) \right|^2 dx + 2 \int_0^1 \left| \int_0^x \gamma_1 v_0(s) ds \right|^2 dx \\ &\leq 4 \int_0^1 |e^{k\gamma_2 x} w(0, \gamma_2 x)|^2 dx + 4 \int_0^1 \left| \int_0^{\gamma_2 x} e^{ks} v(0, s) ds \right|^2 dx + 2 \int_0^1 \left| \int_0^x \gamma_1 v_0(s) ds \right|^2 dx \\ &\leq 4 \int_0^1 |e^{k\gamma_2 x} w(0, \gamma_2 x)|^2 dx + 4K_2 \int_0^1 \int_0^{\gamma_2 x} |v(0, s)|^2 ds dx + 2\gamma_1^2 \int_0^1 \int_0^x |v_0(s)|^2 ds dx \\ &\leq 4K_3 \int_0^{\gamma_2} |w(0, t)|^2 dt + \frac{8K_0 K_1 K_2}{C_0^2 C_1} \int_0^{\gamma_2} [|w_t(0, t)|^2 + k^2 |w(0, t)|^2] dt \\ &\quad + \frac{2\gamma_1^2}{C_0^2 C_2} \int_{\gamma_2}^{\gamma_1 + \gamma_2} [|w_t(0, t)|^2 + k^2 |w(0, t)|^2] dt, \end{aligned} \quad (2.19)$$

where

$$K_2 = \begin{cases} \gamma_2, & \text{if } k = 0, \\ \frac{1}{2k}(e^{2k\gamma_2} - 1), & \text{if } k \neq 0, \end{cases} \quad K_3 = \gamma_1 \max\{1, e^{2\gamma_2 k}\}.$$

Let $y_{\text{obs}}(0, t) = w(0, t) \equiv 0$ in $H^1(0, T)$, then $\frac{d}{dt}y_{\text{obs}}(0, t) = w_t(0, t) = 0$ for a.e. $t \in (0, T)$. Based on (2.14) and (2.19), one can see that $w_0(x) = w_1(x) = 0$ for $x \in (0, 1)$. Thus system (1.1) is approximately observable on $[0, T]$ at the boundary $x = 0$. This completes the proof of Theorem 2.1. $\square$

Remark 2.2. It can be seen from Theorem 2.1 that a same sufficient condition is given to guarantee the exact observability and approximate observability of system (1.1), while the former requires both the displacement and velocity measurement signals at the boundary $x = 0$ and the latter only needs the displacement signal at the boundary $x = 0$. This is in accordance with the fact that the exact observability property is stronger than the approximate observability one.

3. State observer design for system (1.1)

Motivated by the approximate observability of system (1.1) ensured by (ii) of Theorem 2.1, in this section one aims to design a PDE state observer by using the output measurement $w(0, t)$, and a sufficient condition to guarantee the convergence of the estimation error is given. Specially, one constructs a PDE state observer for system (1.1) by using the boundary displacement measurement $w(0, t)$ as follows

$$\begin{cases} z_{tt} - z_{xx} + \gamma z_{xt} + 2kz_t + k\gamma z_x + k^2z = 0, & x \in (0, 1), t > 0, \\ z_x(0, t) = -\alpha[z_t(0, t) + kz(0, t)] + \beta[z_t(0, t) + kz(0, t)]^3, & t \geq 0, \\ z_x(1, t) = Y_x(1, t), & t \geq 0, \\ Y_t(x, t) + \gamma_2 Y_x(x, t) + kY(x, t) = 0, & x \in (0, 1), t > 0, \\ Y(0, t) = c_0(w(0, t) - z(0, t)), & t \geq 0, \\ z(x, 0) = z_0(x), z_t(x, 0) = z_1(x), Y(x, 0) = Y_0(x), & x \in [0, 1], \end{cases} \quad (3.1)$$

where $\gamma, k \in \mathbb{R}$, $\beta > 0$, $\alpha < \gamma_1$, γ_1 and γ_2 are defined as those in (2.2), and $c_0 \neq -1 (\in \mathbb{R})$ is the observer gain. Then, a direct computation shows that the estimation error $p(x, t) = w(x, t) - z(x, t)$ is governed by

$$\begin{cases} p_{tt} - p_{xx} + \gamma p_{xt} + 2kp_t + k\gamma p_x + k^2p = 0, & x \in (0, 1), t > 0, \\ p_x(0, t) = \rho(t)[p_t(0, t) + kp(0, t)], & t \geq 0, \\ p_x(1, t) = -Y_x(1, t), & t \geq 0, \\ Y_t(x, t) + \gamma_2 Y_x(x, t) + kY(x, t) = 0, & x \in (0, 1), t > 0, \\ Y(0, t) = c_0 p(0, t), & t \geq 0, \\ p(x, 0) = p_0(x), p_t(x, 0) = p_1(x), Y(x, 0) = Y_0(x), & x \in [0, 1], \end{cases} \quad (3.2)$$

where

$$\rho(t) = \beta[(z_t(0, t) + kz(0, t))^2 + (z_t(0, t) + kz(0, t))(w_t(0, t) + kw(0, t)) + (w_t(0, t) + kw(0, t))^2] - \alpha. \quad (3.3)$$

According to “Y-part” in (3.2) and the method of characteristics, it is seen that $e^{kt}Y(x, t)$ is constant along each characteristics $x - \gamma_2 t = \text{constant}$, which implies that

$$Y(x, t) = \begin{cases} c_0 e^{-\gamma_1 kx} p(0, t - \gamma_1 x), & t \geq \gamma_1 x, \\ e^{-kt} Y_0(x - \gamma_2 t), & t < \gamma_1 x. \end{cases} \quad (3.4)$$

It is worthy of note that the auxiliary Y-subsystem constructed in the PDE state observer (3.1) plays an important role in the correction of the output measurement estimation error. Moreover, the solution of the auxiliary system can be easily obtained by the method of characteristics, which contributes to the convergence analysis of the estimation error.

Now, one introduces a transformation as follows:

$$\hat{p}(x, t) = [\mathbb{I} + \mathbb{P}_0]p(x, t) = p(x, t) + Y(x, t), \quad \text{for } x \in [0, 1], t \geq 0, \quad (3.5)$$

where its inverse exists because of $\hat{p}(0, t) = (1 + c_0)p(0, t)$ and $Y(x, t) = \mathbb{P}_0 p(x, t)$ satisfies “Y-subsystem” in (3.2). Thus system (3.2) becomes

$$\begin{cases} \hat{p}_{tt} - \hat{p}_{xx} + \gamma \hat{p}_{xt} + 2k\hat{p}_t + k\gamma \hat{p}_x + k^2\hat{p} = 0, & x \in (0, 1), t > 0, \\ \hat{p}_x(0, t) = \frac{\rho(t) - \gamma_1 c_0}{1 + c_0} [\hat{p}_t(0, t) + k\hat{p}(0, t)], & t \geq 0, \\ \hat{p}_x(1, t) = 0, & t \geq 0, \end{cases} \quad (3.6)$$

with the initial state $(\hat{p}_0, \hat{p}_1)$ given by

$$\hat{p}_0(x) = p_0(x) + Y_0(x), \quad \hat{p}_1(x) = p_1(x) - kY_0(x) - \gamma_2 Y_{0x}(x).$$

Consider the space $\mathcal{H} \times H^1(0, 1)$ with the inner product

$$\langle (f_1, g_1, \psi_1), (f_2, g_2, \psi_2) \rangle_{\mathcal{H} \times H^1(0, 1)} = \int_0^1 [f_1' \overline{f_2'} + (g_1 + kf_1)(\overline{g_2 + kf_2}) + \psi_1' \overline{\psi_2'}] dx + \psi_1(0) \overline{\psi_2(0)}.$$

Definition 3.1 ([30]). For given initial data $(p_0, p_1, Y_0) \in \mathcal{H} \times H^1(0, 1)$, if there exist $L > 0$, $\mu > 0$, and $\delta_0 > 0$ such that $\|(p(\cdot, t), p_t(\cdot, t), Y(\cdot, t))\|_{\mathcal{H} \times H^1(0, 1)}^2 \leq L\|(p_0, p_1, Y_0)\|_{\mathcal{H} \times H^1(0, 1)}^2 e^{-\mu t}$ for all $t \geq 0$ whenever $\|(p_0, p_1, Y_0)\|_{\mathcal{H} \times H^1(0, 1)}^2 < \delta_0$, then system (3.2) is said to be exponentially stable; moreover, if $\delta_0 = +\infty$, then system (3.2) is said to be globally exponentially stable.

The globally exponential stability of the resulting estimation error system (3.2), is summarized as the following theorem.

Theorem 3.1. Let $\alpha < \gamma_1$, $\beta > 0$, and $\gamma, k \in \mathbb{R}$. If

$$\text{Condition I: } -1 < c_0 < -\frac{\gamma + 2\alpha}{\gamma_1 + \gamma_2} \text{ and } \gamma_1^2 e^{-(\gamma_1 + \gamma_2)k} < 1 \quad (3.7)$$

holds, then the estimation error system (3.2) is globally exponentially stable, that is, there exist two positive constants L and μ such that

$$\|(p(\cdot, t), p_t(\cdot, t), Y(\cdot, t))\|_{\mathcal{H} \times H^1(0, 1)}^2 \leq L\|(p_0, p_1, Y_0)\|_{\mathcal{H} \times H^1(0, 1)}^2 e^{-\mu t}, \quad (3.8)$$

where L depends on γ, k, c_0 and the initial state $(p_0, p_1, Y_0) \in \mathcal{H} \times H^1(0, 1)$.

Proof. One first shows that system (3.6) is globally exponentially stable. Let $\tilde{u}$ and $\tilde{v}$ be Riemann invariants of (3.6) with

$$\hat{u} = \frac{1}{\gamma_1 + \gamma_2} [\hat{p}_x + \gamma_1(\hat{p}_t + k\hat{p})] \text{ and } \hat{v} = \frac{1}{\gamma_1 + \gamma_2} [\hat{p}_x - \gamma_2(\hat{p}_t + k\hat{p})]. \quad (3.9)$$

It is seen that the left-end boundary condition of (3.6) can be represented as

$$\hat{v}(0, t) = \frac{\rho(t) - (\gamma_1 + \gamma_2)c_0 - \gamma_2}{\rho(t) + \gamma_1} \hat{u}(0, t), \quad t \geq 0,$$

which is well-defined because of $\rho(t) \geq -\alpha$ and $\alpha < \gamma_1$, and the left-end boundary condition of (3.6) can be represented as

$$\tilde{u}(1, t) = -\gamma_1^2 \tilde{v}(1, t), \quad t \geq 0.$$

The initial state $(\hat{u}_0, \hat{v}_0)$ is given by $\hat{u}_0 = \frac{1}{\gamma_1 + \gamma_2} [\hat{p}'_0 + \gamma_1(\hat{p}_1 + k\hat{p}_0)]$ and $\hat{v}_0 = \frac{1}{\gamma_1 + \gamma_2} [\hat{p}'_0 - \gamma_2(\hat{p}_1 + k\hat{p}_0)]$.

Therefore, for $t = (\gamma_1 + \gamma_2)n + \tau$, $n \in \mathbb{N}$, $0 \leq \tau \leq \gamma_1 + \gamma_2$, and $x \in [0, 1]$, the solutions $\hat{u}$ and $\hat{v}$ are expressed as

$$\begin{aligned} \hat{u}(x, t) &= e^{-\gamma_2 k(1-x)} e^{-(\gamma_1 + \gamma_2)kn} (-\gamma_1^2)^n R_n(\tau + \gamma_2 x) \cdots R_1(\tau + \gamma_2 x) (\hat{u}(x, \tau)), \\ \hat{v}(x, t) &= e^{-\gamma_1 kx} e^{-(\gamma_1 + \gamma_2)kn} (-\gamma_1^2)^n R_n(\tau - \gamma_1 x) \cdots R_1(\tau - \gamma_1 x) (\hat{v}(x, \tau)), \end{aligned} \quad (3.10)$$

where

$$\hat{u}(x, \tau) = \begin{cases} e^{-k\tau} \hat{u}_0(h_1(x, \tau) + 1), & 0 \leq \tau \leq \gamma_2(1-x), \\ -\gamma_1^2 e^{-k\tau} \hat{v}_0(-h_2(x, \tau)), & \gamma_2(1-x) < \tau \leq \gamma_2(1-x) + \gamma_1, \\ -\gamma_1^2 e^{-k\tau} R_0(\tau + \gamma_2 x) \hat{u}_0(h_3(x, \tau) + 1), & \gamma_2(1-x) + \gamma_1 < \tau \leq \gamma_1 + \gamma_2, \end{cases} \quad (3.11)$$

$$\hat{v}(x, \tau) = \begin{cases} e^{-k\tau} \hat{v}_0(-h_4(x, \tau)), & 0 \leq \tau \leq \gamma_1 x, \\ e^{-k\tau} R_0(\tau - \gamma_1 x) \hat{u}_0(h_5(x, \tau) + 1), & \gamma_1 x < \tau \leq \gamma_1 x + \gamma_2, \\ -\gamma_1^2 e^{-k\tau} R_0(\tau - \gamma_1 x) \hat{v}_0(-h_6(x, \tau)), & \gamma_1 x + \gamma_2 < \tau \leq \gamma_1 + \gamma_2, \end{cases} \quad (3.12)$$

with $h_i(x, \tau)$, $i = 1, \dots, 6$ given by (2.10) and

$$R_m(\xi) = \frac{\rho(\xi + (\gamma_1 + \gamma_2)(m-1)) - (\gamma_1 + \gamma_2)c_0 - \gamma_2}{\rho(\xi + (\gamma_1 + \gamma_2)(m-1)) + \gamma_1}, \quad m = 0, 1, 2, \dots, n.$$

By $\rho(t) > -\alpha$ and $c_0 \neq -1$, a straightforward computation shows that

$$\begin{cases} R_m(\xi) \in \left(\frac{-\alpha - (\gamma_1 + \gamma_2)c_0 - \gamma_2}{\gamma_1 - \alpha}, 1 \right), & \text{if } c_0 > -1, \\ R_m(\xi) \in \left(1, \frac{-\alpha - (\gamma_1 + \gamma_2)c_0 - \gamma_2}{\gamma_1 - \alpha} \right), & \text{if } c_0 < -1. \end{cases}$$

It is seen that

$$|R_m(\xi)| < 1, \quad (3.13)$$

when $-1 < c_0 < -\frac{2\alpha + \gamma}{\gamma_1 + \gamma_2}$ implying $\frac{-\alpha - (\gamma_1 + \gamma_2)c_0 - \gamma_2}{\gamma_1 - \alpha} > -1$.

For any initial state $(\hat{p}_0, \hat{p}_1) \in \mathcal{H}$, it follows from (3.9) that

$$\|\hat{u}_0\|_{L^2(0, 1)}^2 \leq \frac{2 \max\{1, \gamma_1^2\}}{(\gamma_1 + \gamma_2)^2} \|(\hat{p}_0, \hat{p}_1)\|_{\mathcal{H}}, \quad \|\hat{v}_0\|_{L^2(0, 1)}^2 \leq \frac{2 \max\{1, \gamma_2^2\}}{(\gamma_1 + \gamma_2)^2} \|(\hat{p}_0, \hat{p}_1)\|_{\mathcal{H}}. \quad (3.14)$$

Consequently, by using (3.10)–(3.14), there exists a constant $M > 0$ such that

$$\max\{\|\hat{u}(\cdot, t)\|_{L^2(0,1)}^2, \|\hat{v}(\cdot, t)\|_{L^2(0,1)}^2\} \leq M[\gamma_1^2 e^{-(\gamma_1+\gamma_2)k}]^{2n},$$

when $-1 < c_0 < -\frac{2\alpha+\gamma}{\gamma_1+\gamma_2}$. This, together with the **Condition I** in (3.7) and

$$\begin{aligned} \int_0^1 [|\hat{p}_x(x, t)|^2 + |\hat{p}_t(x, t) + k\hat{p}(x, t)|^2] dx &= \int_0^1 [|\gamma_2 \hat{u}(x, t) + \gamma_1 \hat{v}(x, t)|^2 + |\hat{u}(x, t) - \hat{v}(x, t)|^2] dx \\ &= \int_0^1 [(\gamma_2^2 + 1)|\hat{u}(x, t)|^2 + (\gamma_1^2 + 1)|\hat{v}(x, t)|^2] dx, \end{aligned}$$

implies that $\|\hat{p}(\cdot, t), \hat{p}_t(\cdot, t)\|_{\mathcal{H}}$ is exponentially stable, that is, there exist two positive constants L_0 and μ such that

$$\|\hat{p}(\cdot, t), \hat{p}_t(\cdot, t)\|_{\mathcal{H}}^2 \leq L_0 e^{-\mu t}, \quad (3.15)$$

where L_0 depends on γ, k and the initial state $(\hat{p}_0, \hat{p}_1)$, and μ depends on γ, k .

Next, it suffices to prove that $\|Y(\cdot, t), Y_t(\cdot, t)\|_{\mathcal{H}}$ is exponentially stable. Let

$$\Psi(t) = \psi(t) + 2k \int_0^t \psi(s) ds,$$

where

$$\psi(t) = \int_0^1 (x-1)[2\hat{p}_x(x, t)(\hat{p}_t(x, t) + k\hat{p}(x, t)) + \gamma \hat{p}_x^2(x, t)] dx$$

with $|\psi(t)| \leq (1+|\gamma|)\|(\hat{p}(\cdot, t), \hat{p}_t(\cdot, t))\|_{\mathcal{H}}^2$. Differentiate $\Psi(t)$ along system (3.6) to obtain

$$\begin{aligned} \dot{\Psi}(t) &= \dot{\psi}(t) + 2k\psi(t) \\ &= |\hat{p}_t(0, t) + k\hat{p}(0, t)|^2 + |\hat{p}_x(0, t)|^2 - \int_0^1 [|\hat{p}_t(x, t) + k\hat{p}(x, t)|^2 + |\hat{p}_x(x, t)|^2] dx \\ &\geq |\hat{p}_t(0, t) + k\hat{p}(0, t)|^2 - \|(\hat{p}(\cdot, t), \hat{p}_t(\cdot, t))\|_{\mathcal{H}}^2, \end{aligned}$$

which, combining with (3.15), yields that

$$\begin{aligned} \int_{t-\gamma_1}^t |\hat{p}_t(0, s) + k\hat{p}(0, s)|^2 ds &\leq \Psi(t) - \Psi(t - \gamma_1) + \int_{t-\gamma_1}^t \|(\hat{p}(\cdot, s), \hat{p}_t(\cdot, s))\|_{\mathcal{H}}^2 ds \\ &= \psi(t) - \psi(t - \gamma_1) + 2k \int_{t-\gamma_1}^t \psi(s) ds + \int_{t-\gamma_1}^t \|(\hat{p}(\cdot, s), \hat{p}_t(\cdot, s))\|_{\mathcal{H}}^2 ds \\ &\leq (1+|\gamma|)\|(\hat{p}(\cdot, t), \hat{p}_t(\cdot, t))\|_{\mathcal{H}}^2 + (1+|\gamma|)\|(\hat{p}(\cdot, t - \gamma_1), \hat{p}_t(\cdot, t - \gamma_1))\|_{\mathcal{H}}^2 \\ &\quad + [2|k|(1+|\gamma|) + 1] \int_{t-\gamma_1}^t \|(\hat{p}(\cdot, s), \hat{p}_t(\cdot, s))\|_{\mathcal{H}}^2 ds \\ &\leq L_0^* e^{-\mu t}, \end{aligned} \quad (3.16)$$

where

$$L_0^* = L_0 \left\{ (1+|\gamma|)(1+e^{\gamma_1\mu}) + [2|k|(1+|\gamma|) + 1] \frac{e^{\gamma_1\mu} - 1}{\mu} \right\}.$$

Then, by (3.4), $\hat{p}(0, t) = (1+c_0)p(0, t)$, and (3.16), one obtains that

$$\begin{aligned} \int_0^1 |Y_x(x, t)|^2 dx &= \frac{\gamma_1^2 c_0^2}{(1-c_0)^2} \int_0^1 e^{-2\gamma_1 kx} |\hat{p}_t(0, t - \gamma_1 x) + k\hat{p}(0, t - \gamma_1 x)|^2 dx \\ &\leq \max\{1, e^{-2\gamma_1 k}\} \frac{\gamma_1^2 c_0^2}{(1-c_0)^2} \int_{t-\gamma_1}^t |\hat{p}_t(0, s) + k\hat{p}(0, s)|^2 ds \\ &\leq \widehat{L}_0^* e^{-\mu t}, \end{aligned} \quad (3.17)$$

where $\widehat{L}_0^* = \max\{1, e^{-2\gamma_1 k}\} \frac{\gamma_1^2 c_0^2}{(1-c_0)^2} L_0^*$. Similarly, one also has

$$\int_0^1 |Y_t(x, t) + kY(x, t)|^2 dx = \frac{c_0^2}{(1-c_0)^2} \int_0^1 e^{-2\gamma_1 kx} |\hat{p}_t(0, t - \gamma_1 x) + k\hat{p}(0, t - \gamma_1 x)|^2 dx \leq \widetilde{L}_0^* e^{-\mu t}$$

with $\tilde{I}_0^* = \gamma_2^2 \hat{I}_0^*$, which, combining with (3.5), (3.15), and (3.17), implies that

$$\begin{aligned} \|(p(\cdot, t), p_t(\cdot, t))\|_{\mathcal{H}}^2 &\leq 2 \int_0^1 [|\hat{p}_x(x, t)|^2 + |\hat{p}_t(x, t) + k\hat{p}(x, t)|^2] dx \\ &\quad + 2 \int_0^1 [|\hat{Y}_x(x, t)|^2 + |\hat{Y}_t(x, t) + k\hat{Y}(x, t)|^2] dx \\ &\leq L_1 e^{-\mu t}, \end{aligned} \quad (3.18)$$

where $L_1 = 2L_0 + 2\hat{I}_0^* + 2\tilde{I}_0^*$. By $Y(0, t) = c_0 p(0, t)$ given by (3.2) and the Sobolev trace-embedding theorem, it is seen that

$$|Y(0, t)| = |c_0| |p(0, t)| \leq |c_0| \|(p(\cdot, t), p_t(\cdot, t))\|_{\mathcal{H}}. \quad (3.19)$$

Therefore, (3.8) is obtained from (3.17), (3.18) and (3.19) with $L = (c_0^2 + 1)L_1 + \hat{I}_0^*$. Theorem 3.1 is then completely proved. $\square$

Remark 3.1. It is noteworthy that observability is the important structural property that has close relationship with state observer. Nevertheless, observability, especially for systems with nonlinearity, does not directly means the existence of a convergent state observer. **Condition 1**, which is the sufficient condition of Theorem 3.1 concerning the restriction of systems parameters and observer gain to guarantee the exponential convergence of the designed PDE state observer using only displacement measurement, is required additionally.

Remark 3.2. In practical applications, direct measurement of boundary velocity can be relatively difficult and the associated transducer tends to be both heavy and costly. In contrast to boundary velocity, measuring boundary displacement is a relatively straightforward process (see, e.g., [15]). From the perspective of observer design and practical meaning, it is expected that we can use as few output measurements as possible to design an observer. In this section it is shown with a rigorous proof that the PDE state observer can be designed with exponential convergence by using the boundary displacement measurement $y_{obs}(t) = w(0, t)$ only, corresponding to the approximate observability. The boundary output measurement $\mathbf{y}_{obs}(t) = (w(0, t), w_t(0, t))^T$ can certainly be utilized to design an observer for system (1.1) through the same methodology as presented in this section, but this is unnecessary as stated above.

4. Numerical simulations

In this section, the state estimation effect for a PDE system described by (1.1) is validated by numerical simulations. The numerical results are programmed by MATLAB, and the time and spatial steps are both taken as 0.02. One performs the numerical simulations via analytic solutions (2.8)–(2.9) and (3.10)–(3.12). The initial values $(w_0, w_1) \in \mathcal{H}$ of system (1.1) are chosen to be

$$w_0(x) = (x - 0.5)^3 \sin^4(\pi x), \quad w_1(x) = 0, \quad x \in [0, 1],$$

and the initial values (z_0, z_1, Y_0) of the PDE state observer (3.1) are given by

$$z_0(x) = -w_0(x), \quad z_1(x) = 0, \quad Y_0(x) = 0, \quad x \in [0, 1].$$

Some concerning parameters are chosen as

$$\gamma_1 = 0.5, \quad \gamma_2 = 2, \quad k = 0.01, \quad \alpha = 0.01, \quad \beta = 1, \quad c_0 = -0.5.$$

It is seen that all the conditions in Theorem 3.1 are satisfied. Therefore, it follows from Theorem 3.1 that the estimation error system (3.2) is exponentially stable. In order to compare the result, Fig. 1 displays the profiles of w_x , w_t , and w of system (1.1) with $t \in [0, 15]$, and Fig. 2 displays the profiles of p_x , p_t , and p of the error system (3.2) with $t \in [0, 15]$. For $x = 0.3$, the profiles of (w_x, z_x, p_x) , (w_t, z_t, p_t) , and (w, z, p) with $t \in [0, 12]$ are displayed in Fig. 3. It can be observed from Figs. 2 and 3 that the state estimation effect of the proposed PDE state observer is quite satisfactory.

5. Concluding remarks

This paper investigates observability and state observer design for a class of one-dimensional hyperbolic PDE systems with van de Pol type nonlinear boundary condition, where the systems can exhibit complex dynamics due to its imbalance of energy flows. Firstly, for different boundary output measurements, exact observability and approximate observability of the systems are respectively obtained by adopting approaches of characteristics and boundary nonlinear reflections. Secondly, under approximate observability of the systems with boundary displacement measurement only, an infinite-dimensional state observer using the boundary measurement is designed, and the exponential convergence of the estimation error can be obtained provided that an additional condition is satisfied. Rigorous theoretical proofs are given for the main results of the paper, with the estimation effect of the proposed state observer be validated by some simulations. There are two interesting issues deserved to be explored further. One is the development of the observability and PDE state observer design to multi-dimensional hyperbolic PDE systems. Another potential problem is the boundary output feedback stabilization or output tracking of the hyperbolic PDE systems based on the PDE state observer.

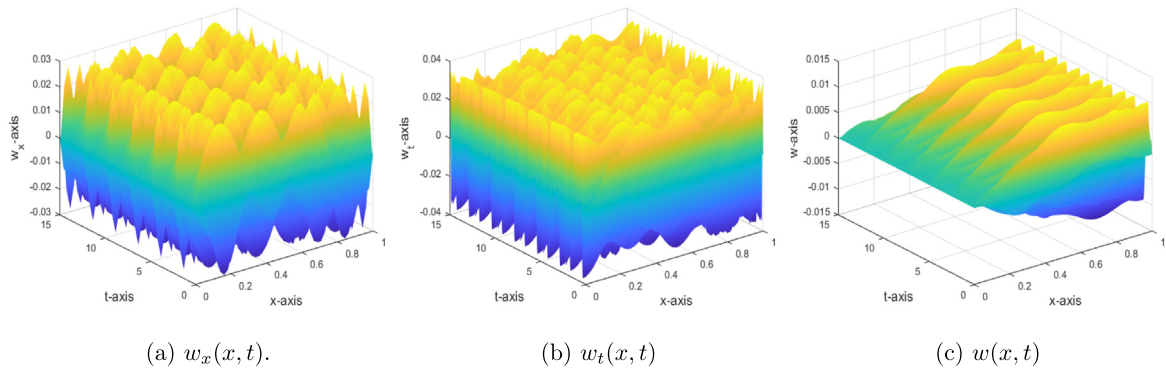


Fig. 1. The profiles of w_x , w_t , and w for system (1.1) with $t \in [0, 15]$.

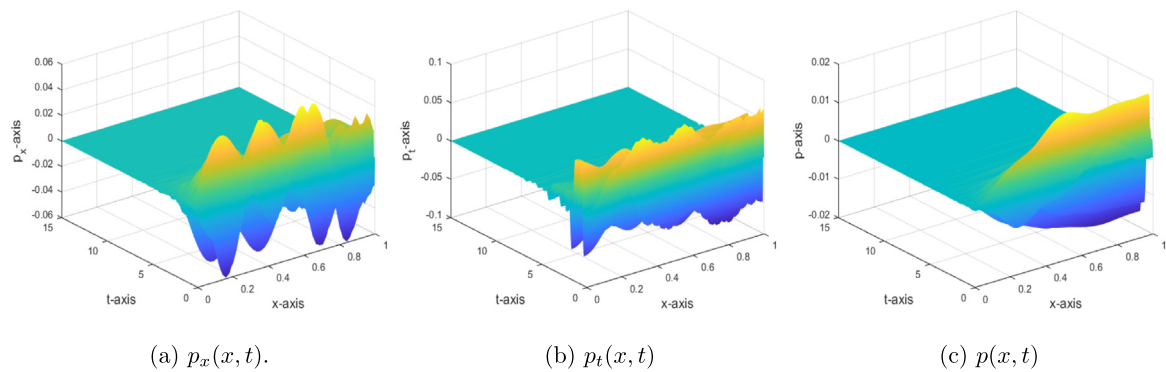


Fig. 2. The profiles of p_x , p_t , and p for system (3.2) with $t \in [0, 15]$.

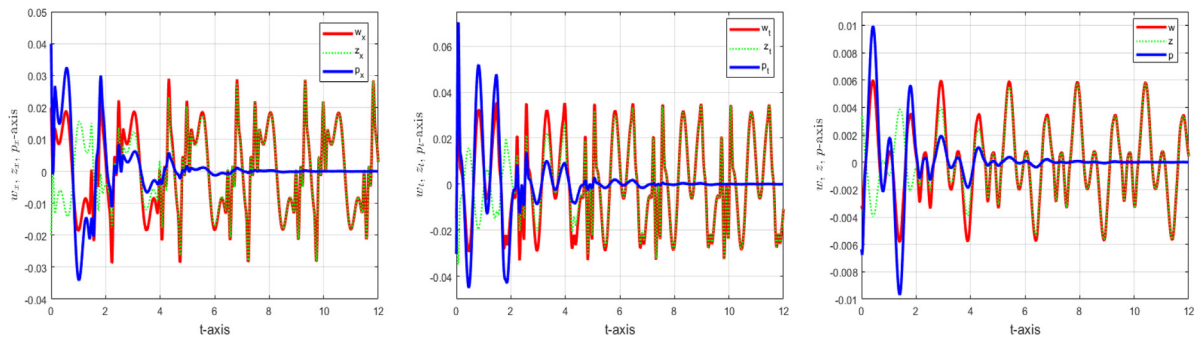


Fig. 3. The profiles of (w_x, z_x, p_x) , (w_t, z_t, p_t) , and (w, z, p) with $x = 0.3$ and $t \in [0, 12]$.

CRedit authorship contribution statement

Qiaomin Xiang: Investigation, Conceptualization, Writing – original draft. **Ze-Hao Wu:** Writing – review & editing, Supervision. **Feiqi Deng:** Supervision, Methodology. **Chufen Wu:** Visualization.

Declaration of competing interest

The authors declare that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Data availability

No data was used for the research described in the article.

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